AYAN MITRA

Research Scientist | Scientific Software and Agentic AI for Research

Champaign, Illinois, USA | ayan@illinois.edu | +1 (217) 974 0644

LinkedIn | GitHub | ORCID 0000 0002 9436 8871

PROFILE

Research scientist and scientific software developer with a PhD in physics and experience building reproducible computational systems for data intensive science. Leading contributor to SNANA and pipeline scientist for the Vera C. Rubin Observatory LSST Dark Energy Science Collaboration. Builds Python and high performance computing workflows across simulation, machine learning, uncertainty, and inference at million sample scale. Created an agentic operations assistant for scientific pipelines with source grounded diagnoses, scoped tool permissions, and regression evaluations. Interested in open infrastructure for AI assisted science that preserves provenance, reproducibility, and researcher judgment.

TECHNICAL STRENGTHS

- **Agentic AI and evaluation:** Retrieval augmented assistants, Claude Code skills, tool integration, evaluation suites, source grounded responses, permission design, local and hosted model support.
- **Scientific workflows and HPC:** Python, NumPy, SciPy, Bash, NERSC Perlmutter, SLURM, containerization, orchestration, debugging, and automated workflows processing millions of simulated events.
- **Research engineering:** Git, GitHub, continuous integration, testing, code review, documentation, reproducible releases, R, Fortran, and C.
- **Machine learning and inference:** Classification, neural simulation based inference, scikit learn, SCONE, Bayesian inference, Monte Carlo simulation, uncertainty quantification, and robustness testing.

SELECTED AGENTIC AI AND SOFTWARE PROJECTS

SNANA Pipeline Assistant: Independent prototype for scientific HPC operations. A Claude Code skill and standalone command line interface diagnose stale locks, configuration mismatches, memory failures, and scheduler state using a curated knowledge base and SLURM tools. Answers trace to sources. Tool access is read only by default, with one scoped, non destructive write action. A 20 case regression suite gates changes. Supports hosted providers and local models for privacy sensitive settings. Repository

Firecrown Wrapper: Open source Python command line workflow connecting Firecrown and CosmoSIS for reproducible cosmology inference. Includes tests, continuous integration, structured outputs, and batch use on NERSC systems. Repository

Gravitational wave equation of state classifier: Runnable 1D convolutional neural network and interactive Gradio demo for scientific classification. The reproduction run reaches 86.5 percent held out accuracy and macro F1 of 0.86. Repository

SCONE Tools: Data extraction and visualization tools for neural supernova classifier outputs, including indexed lookup across millions of classified events at NERSC scale. Repository

RESEARCH AND ENGINEERING EXPERIENCE

CAPS Fellow and Pipeline Scientist | NCSA, UIUC and LSST DESC | December 2023 to present

- Lead development and validation of a research pipeline connecting simulation, feature extraction, machine learning classification, bias correction, uncertainty propagation, and statistical inference.
- Design and run large scale experiments on NERSC and SLURM, diagnose multistage failures, and validate scientific behavior across millions of simulated events.
- Serve as a leading contributor to SNANA, implementing host galaxy redshift uncertainty as a fitted parameter and propagating probability information through downstream analyses.
- Integrate Firecrown likelihood infrastructure, bridge heterogeneous Rubin data products into reproducible workflows, and collaborate across technical working groups and internal reviews.
- Mentor researchers at UIUC, Cambridge, Stanford, and Harvard on machine learning classification, inference, and pipeline development.

Rubin Postdoctoral Fellow | IUCAA, India | 2021 to 2023

- Developed photometric classification and redshift systematics research for Rubin scale analyses, translating methods into validated collaboration software and publications.
- Co led studies of rare signal classification using simulated gravitational wave detector data.

Postdoctoral Fellow | Energy Cosmos Laboratory, Nazarbayev University | 2019 to 2020

- Conducted statistical and computational research across cosmology and gravitational wave inference, developed portable analysis workflows, and supervised student research.

SELECTED RESEARCH

- FlowSN, neural simulation based inference under realistic selection effects for supernova cosmology, coauthor, arXiv 2603.11165, 2026.
- Evaluating machine learning models for supernova gravitational wave signal classification, coauthor, Machine Learning: Science and Technology 5, 045 077, 2025.
- Probing nuclear physics with supernova gravitational waves and machine learning, first author, MNRAS 529, 3582 to 3592, 2024.
- Dark energy reconstruction with artificial neural networks on simulated Rubin Observatory supernova data, first author, arXiv 2402.18124, 2024.
- Fully photometric Type Ia supernova inference combining host galaxy redshift uncertainty and learned classification, arXiv 2512.06319, 2025.

EDUCATION

PhD, Supernova Cosmology, Université Pierre et Marie Curie, Sorbonne Université, France, 2013 to 2016.

MSc, Cosmology, University of Sussex, 2011 to 2013.

BSc with Honors, Physics, St. Xavier's College, University of Calcutta, 2008 to 2011.

LEADERSHIP, RECOGNITION, AND LANGUAGES

- LSST DESC Builder and internal reviewer. Journal referee for Nature Communications, MNRAS, and ApJ. NOIRLab Time Allocation Committee panel member for 2026B.
- CAPS Fellowship, Rubin Postdoctoral Fellowship, ACME TNA, Lagrange Fellowship at CNRS, and Data Incubator Fellowship.
- English and Bengali native. Russian advanced. French B2. Hindi working.